PI 5 POE NVME CASE

A slide-in tube for a Raspberry Pi 5 with the **Waveshare PoE HAT (G)** on its header and a **Pimoroni NVMe Base Duo** underneath. No screws or standoffs in either board: **each board rests on its own ledge**, the PCIe ribbon is the only link between them, a **TPU card retainer** springs them against the closed USB end, and a **click-in PLA lid** closes the SD end against the closed USB end. The body prints standing on its USB end, so the ledges, vents and ports need no supports.

PoE HAT (G) · WS-27670 | NVMe Base Duo · PIM704 | Pi 5 + Active Cooler | PETG or PLA body · PLA lid · TPU feet | Prusa MK3S+

OUTSIDE L × W × H	HEIGHT ON FEET	PARTS	PRINT, ALL PARTS
106.3 × 68.8 × 64.6	70	body · retainer · lid · 4 feet or DIN bracket	5h 48m

Print test, 11 Nov

- **Body: printed well** (PLA, 10 Nov). The ledges hold the NVMe Base Duo; you shook the case and the base stayed put. No change to the body, so no reprint.
- **TPU lid: the snap bumps came out as faint nubs.** The slicer did include them, but they were only 0.3–0.8 mm proud over five 0.3 mm layers, which TPU smoothed away. Even printed perfectly, only 0.55 mm would have reached into the windows.
- **v0.3 lid: PLA, with flexing latch arms** (your suggestion). One arm per long side, cut free by slots and fixed at the top. Its 1 mm hook clicks into the upper side window that's already in your body. The toolpath now shows the full 1.0 mm hook over 12 layers.
- **v0.4: a TPU card retainer holds the boards; the lid just closes** (your suggestion). The retainer locks in with bumps in the first vent slots, and its fingers spring against the board corners. The lid loses its prongs.
- **v0.4 feet suit your M3 × 8 button heads.** The v0.2 feet would have let the screw bottom out 0.8 mm before it clamped. Now they're 6 mm tall with a 3 mm counterbore.
- **To print: the lid, the retainer and the feet.** The body stays as printed.

Settled

- **SSDs: Samsung 980 series.** Single-sided, 2.38 mm, no heatsink. On Pimoroni's 5 mm SSD standoffs they reach about 7.4 mm below the Duo, inside the 9.5 mm allowed.
- **HAT height: unknown, so kept at 16 mm.** That's the tall case. If your HAT sits lower, you just get more air above it.
- **HAT stays on its headers.** No hold-down ribs.
- **SD tray dropped.** Fit the microSD card before the PCIe ribbon.

The print-ready STLs are attached to this PDF (open the attachments or paperclip panel in your PDF reader). Each is already oriented for the bed:

- `pi5_case_body_PETG.stl · 374 KB`
- `pi5_case_card_retainer_TPU.stl · 26 KB`
- `pi5_case_lid_PLA.stl · 46 KB`
- `pi5_case_feet_x4_TPU.stl · 94 KB`
- `pi5_case_din_bracket_PETG.stl · 133 KB`

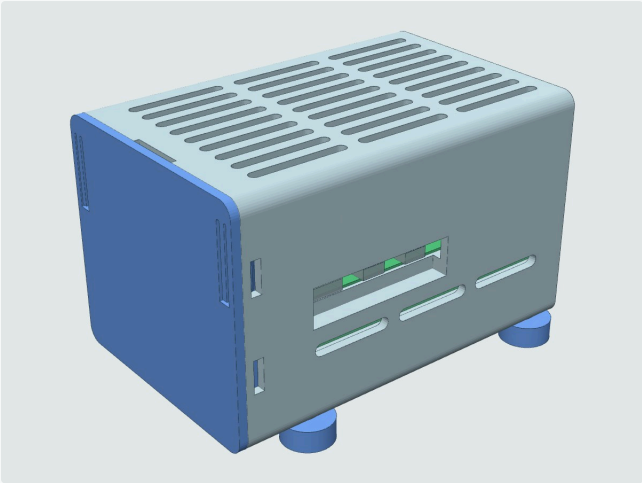
Print time

PRUSASLICER 2.7 · 0.30 MM DRAFT (LID 0.20 MM SPEED) · 0.4 NOZZLE · MK3S+

	Material	Time
Body	PETG, USB end on the bed	**4h 27m** 77 g
Lid	PLA, 0.20 mm, outer face on the bed (PETG: 51m)	**48m** 14 g
Card retainer	TPU, eSun 95A FAST, back face on the bed	**18m** 4 g
Feet × 4	TPU, eSun 95A FAST	**16m** 3 g
DIN bracket	PETG, rail side down. Draft: 1h 39m. Recommended 4 walls, 40% gyroid	**2h 24m** 43 g
Case + feet		**5h 48m**

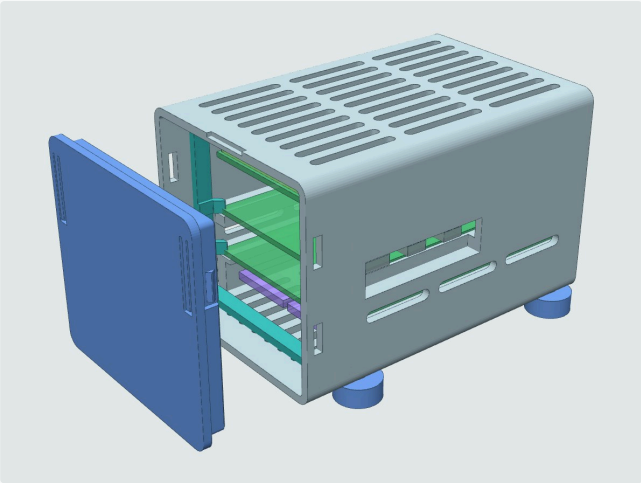
Walls, roof, floor and end are 2.4 mm. The same body with your usual 3 mm walls slices at 5h 23m, so 2.4 saves 56m. The stiffness comes from the two lips and the thick SSD-bay walls, which run the full length on both sides like rails, and from the bars between the vent rows.

3D views



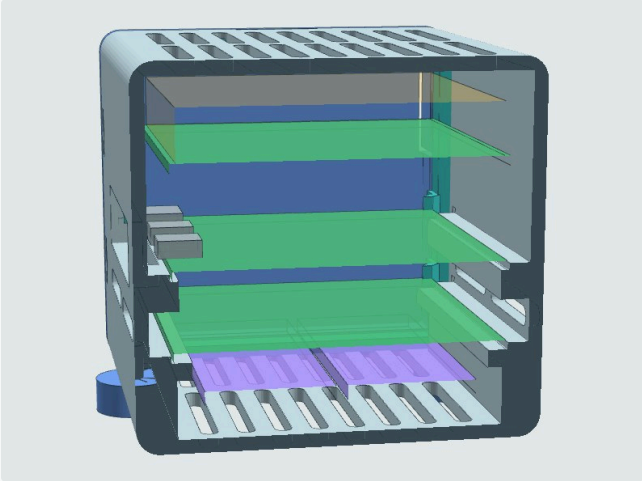
ASSEMBLY

Case on its feet with the boards in place (boards, SSDs and HAT parts shown as blocks).



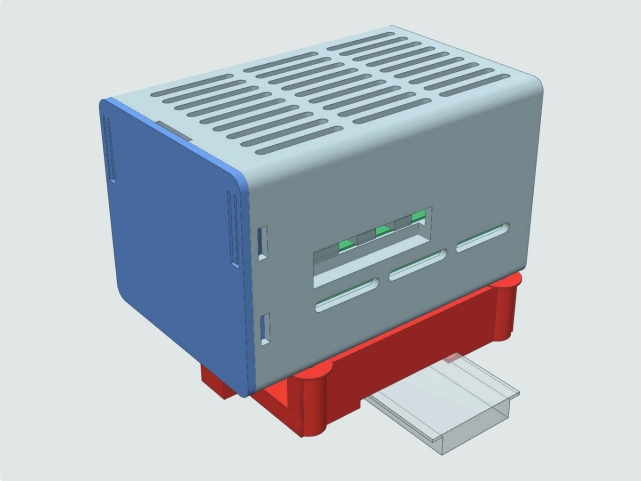
LID OUT

The PLA lid pulled 30 mm off the SD end. The TPU card retainer (teal) stays in, behind the boards.



SECTION THROUGH THE LEDGES

Cut across the case: the Pi on the upper ledge, the NVMe Base Duo on the lower ledge, SSDs in the bay.



ON A DIN RAIL

Feet replaced by the DIN rail bracket, clipped to a TS35 rail.

Rendered from the 3D model. The interactive viewer is a separate page.

Drawings

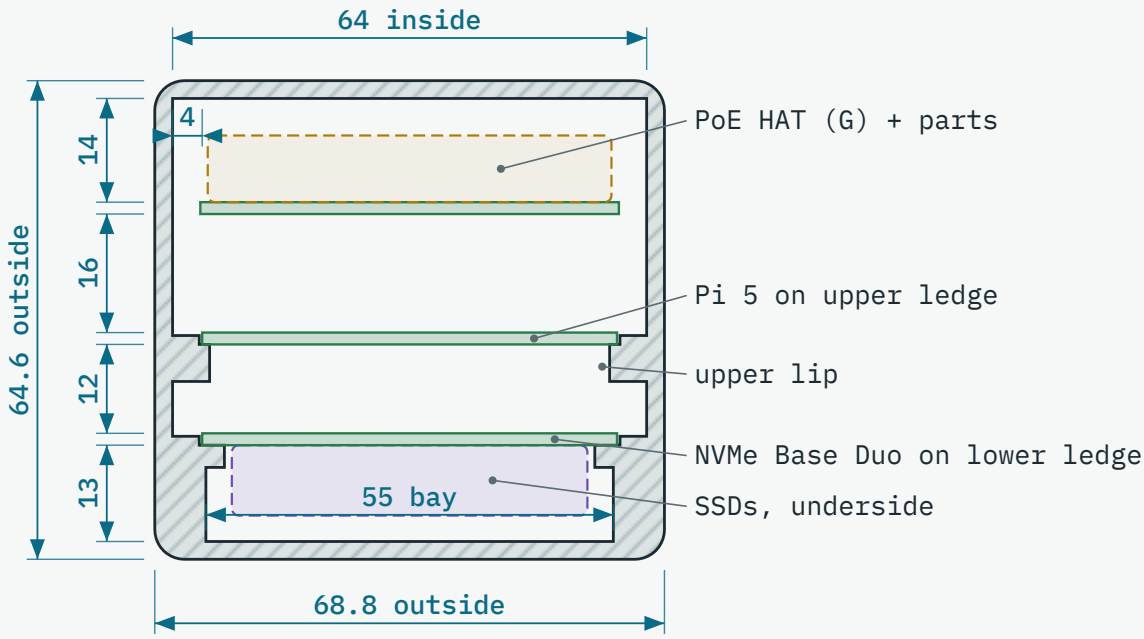
Dimensions in mm. The drawings are not to scale on the page; a scale in a title (3:1, 2:1) is relative to the other views.

body (cut) lid / TPU parts PCB SSD zone HAT parts PCIe ribbon dimension (mm)

CROSS-SECTION

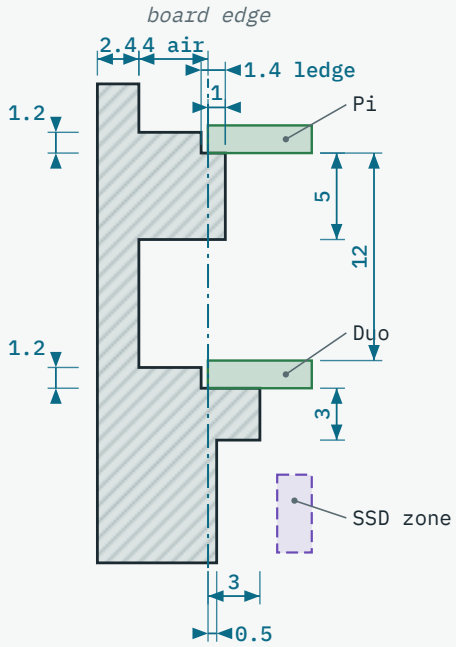
SECTION A–A · ACROSS

Your second sketch. Each board sits on its own ledge and nothing screws into either board. The upper lip runs between the two boards: the Pi rests on its top, and the Duo's edge runs in the recess under it, resting on the lower lip. Below that the wall steps back out to make the SSD bay.



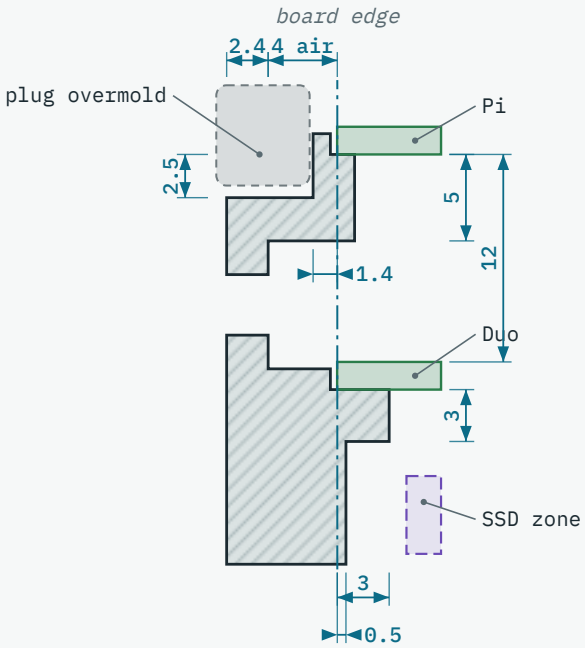
DETAIL · THE LEDGES (3:1)

A 1.2 mm fence on each ledge, lower than the PCB, keeps each board centred with 0.4 mm play a side. The Pi's ledge only reaches 1 mm under it: the GPIO solder joints and the jack tabs on the Pi's underside start about 1.4 mm in from the edge. The Duo's ledge reaches 3 mm, clear of the SSDs.



DETAIL · LEDGES AT THE PORT WINDOW (3:1)

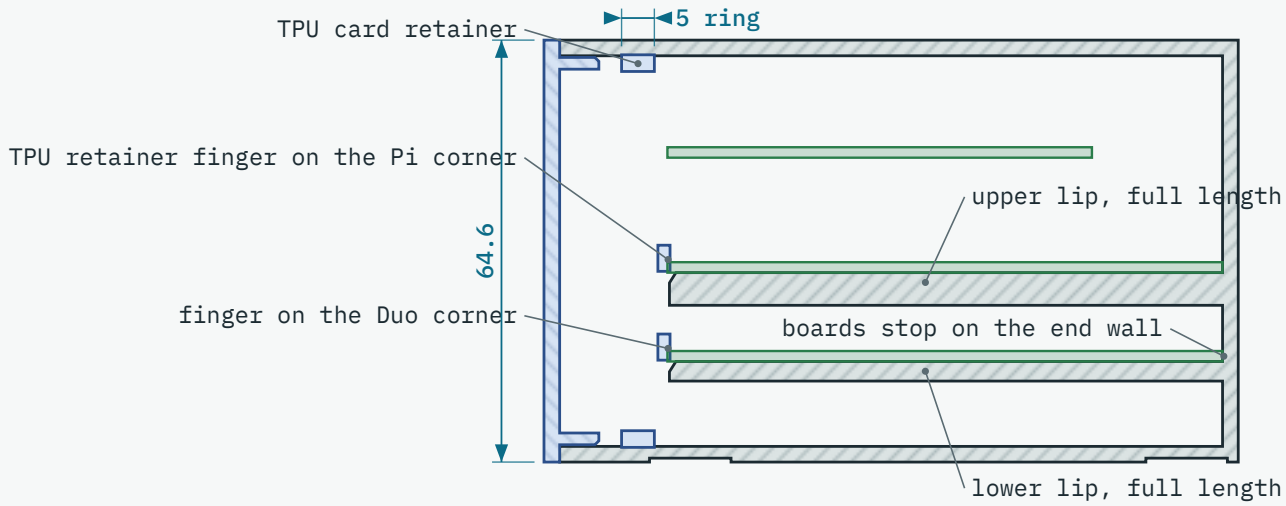
A chunky USB-C plug's overmold hangs about 2 mm below the Pi's board edge. Along the 42 mm port window the upper lip's outer half drops 2.5 mm so the plug can seat; the 1 mm ledge under the Pi carries on, tied to the wall lower down.



LENGTHWISE

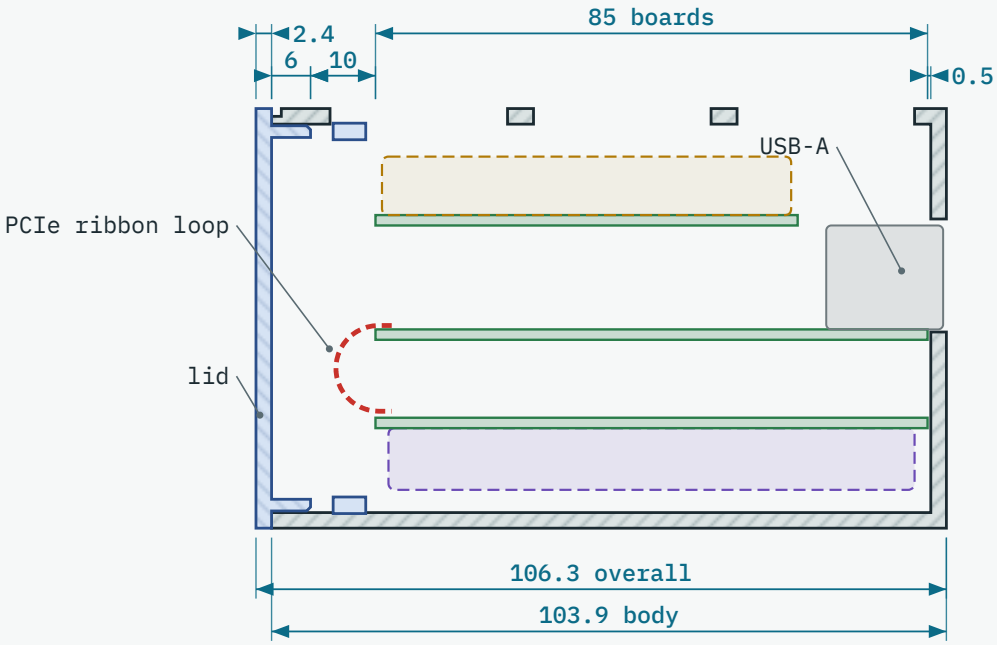
SECTION C–C · ALONG THE GPIO-SIDE LEDGES

Both lips run the full length. The boards slide in from the SD end, USB first, together (the ribbon joins them) and stop on the closed end wall. The TPU card retainer (v0.4) sits behind them; its fingers spring against both boards' corners.



SECTION B–B · DOWN THE MIDDLE

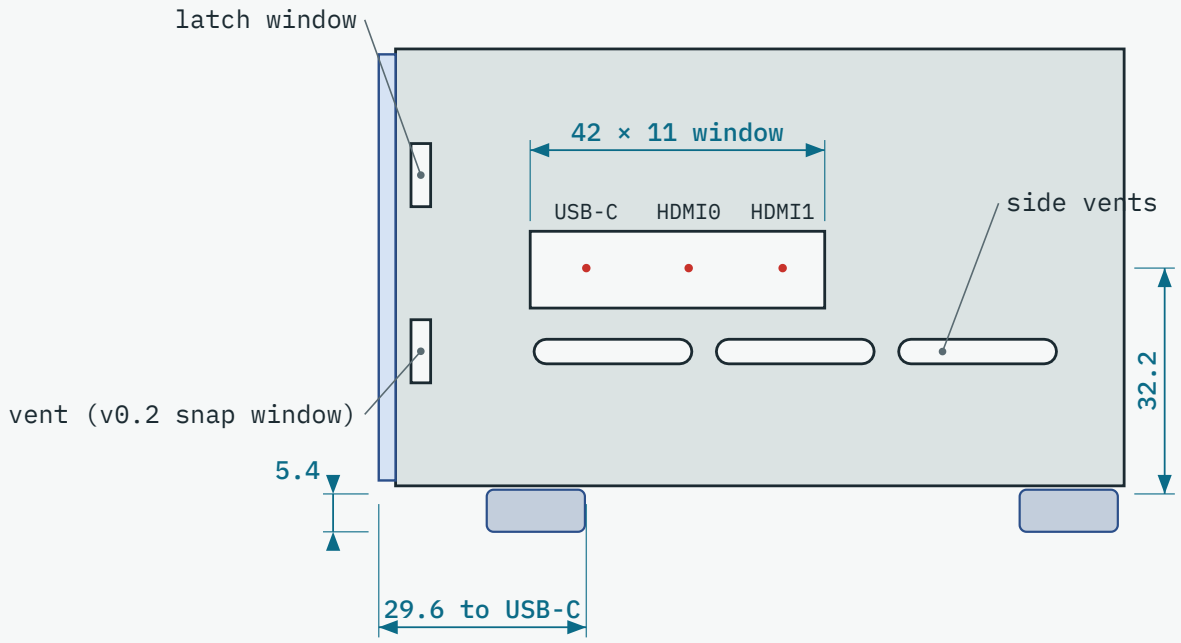
Centre line, through the first USB stack. The PCIe ribbon loops round the SD end of the boards; there is 10 mm for it between the boards and the lid.



OUTSIDE

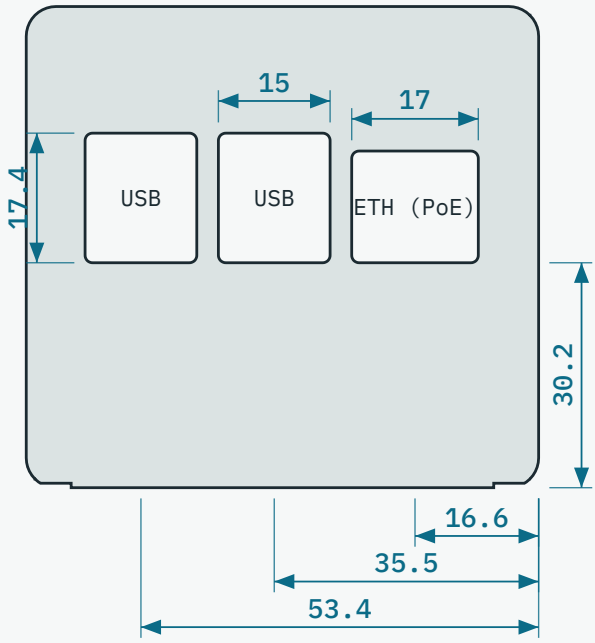
SIDE ELEVATION · HDMI / USB-C SIDE

One 42 × 11 window covers USB-C and both micro HDMI, sized for a dock cable's overmold. Below it, slots vent the space between the two boards. Near the lid, the upper small window takes the lid's latch hook; the lower one is now just a vent. The other long side has the same vents without the port window.



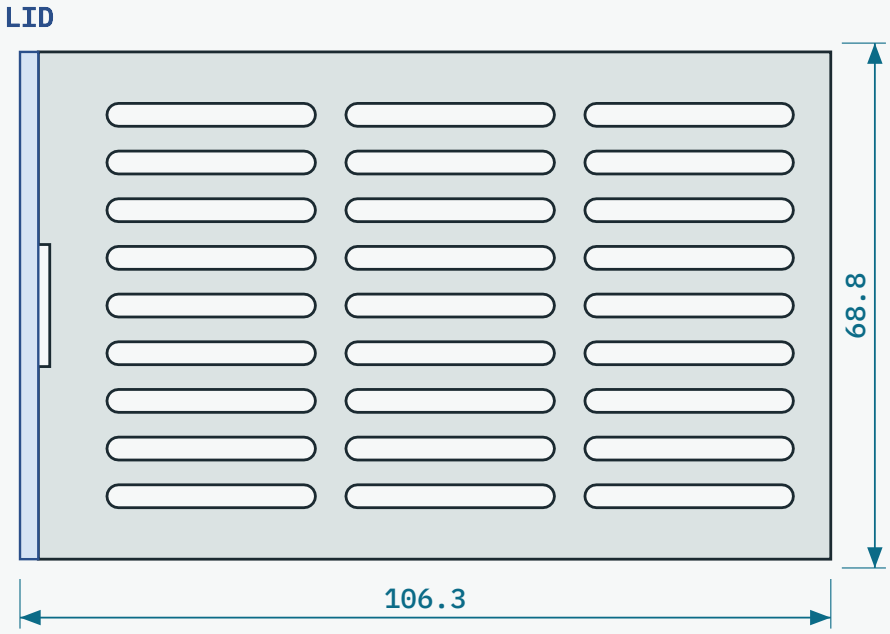
USB / RJ45 END

The closed end of the body. It prints first, flat on the bed. Cut-outs are 0.5 mm clear of each jack.



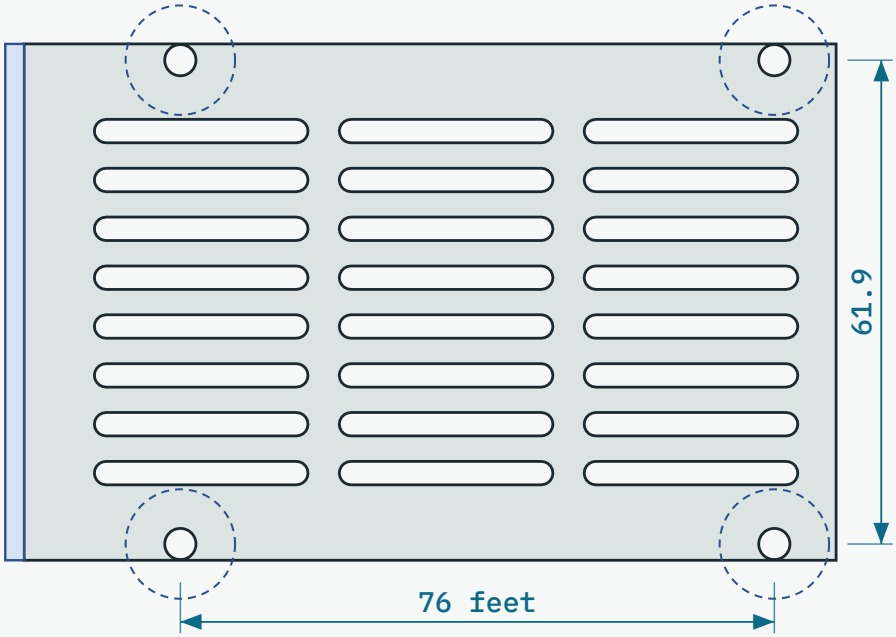
TOP

Roof vents: 3 mm slots on a 6.25 mm pitch in three rows. The bars between rows stiffen the roof.



BOTTOM

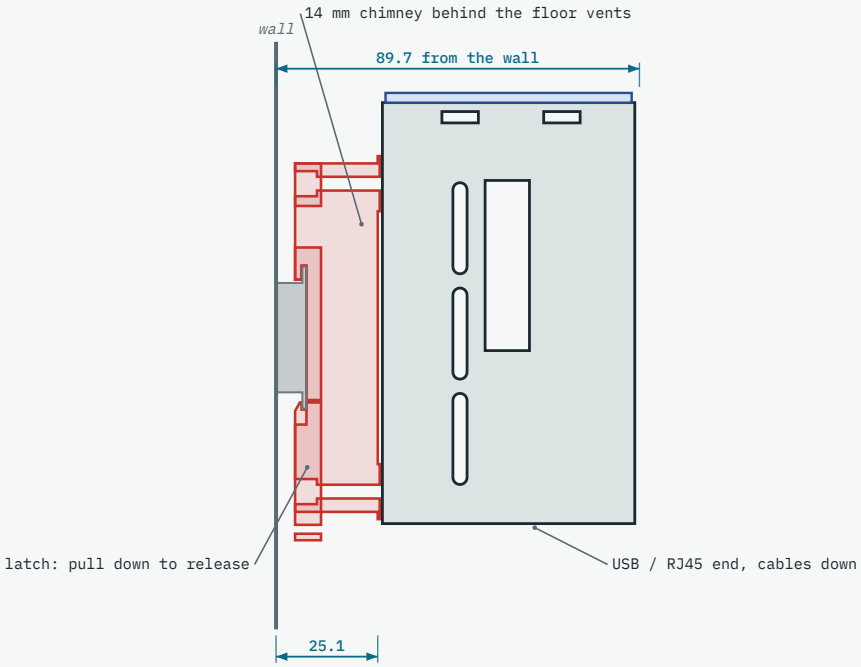
Floor vents under the SSDs. M3 heat-set inserts for the four feet sit in the thick SSD-bay walls, inside 0.6 mm locating recesses.



DIN RAIL BRACKET

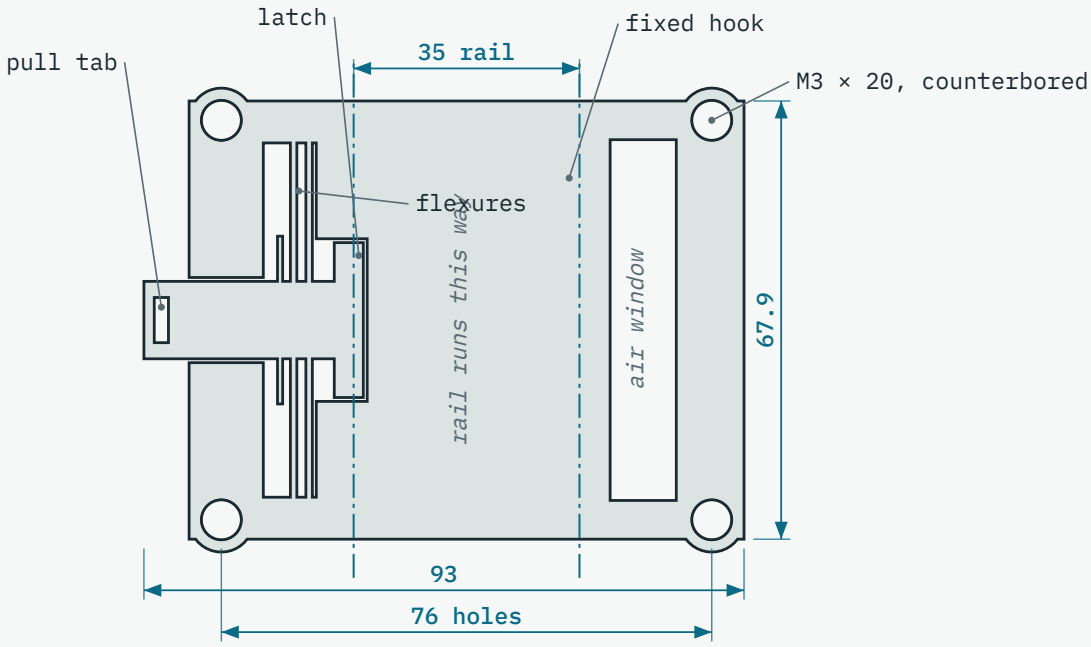
ON THE RAIL · SIDE VIEW

Optional mount in place of the feet. The case stands with its length vertical, here USB end down so cables hang (the bolt pattern is symmetric, so it also bolts on the other way up). Its floor vents open into a 14 mm chimney behind it.



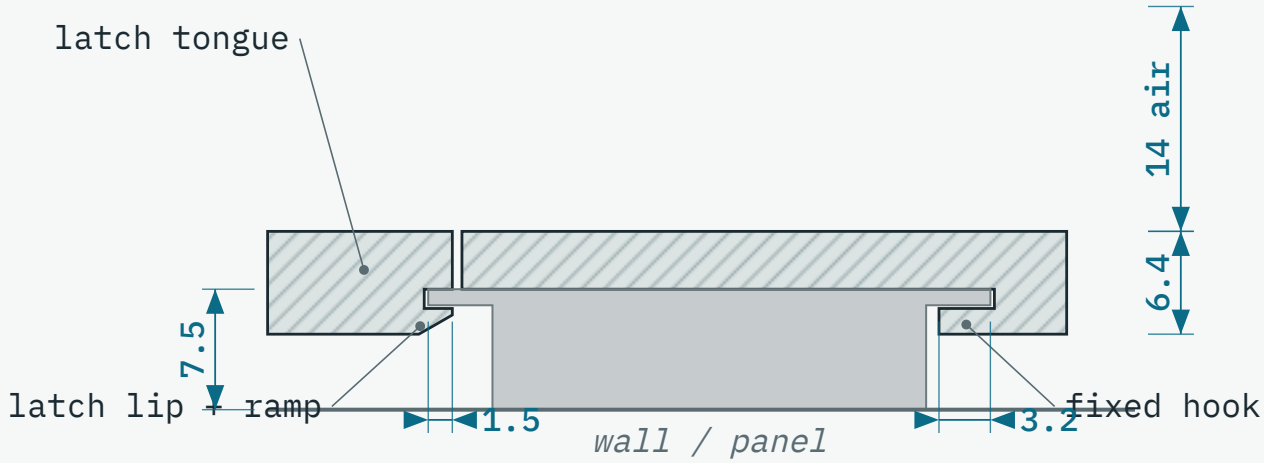
BRACKET · PLAN

Seen from the case side, cut through the plate. Four M3 × 20 socket heads go through counterbored bosses into the foot inserts. The tall side ribs run under the case's thick lower walls, clear of the floor vents, and stiffen the whole bracket. The window and the flexure gaps let air through.



BRACKET · CLIP ON THE RAIL (2:1)

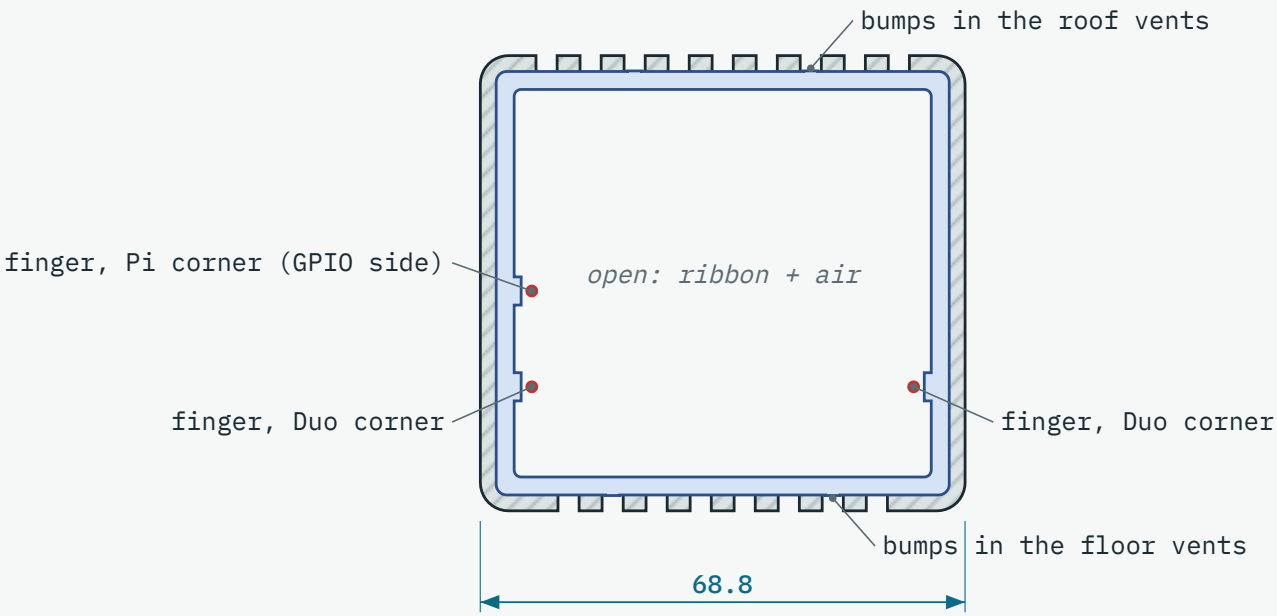
A fixed hook runs the full width on one flange. The other flange gets a latch on four flexures with a lead-in ramp, so it snaps on when pushed. To remove it, lever the pull tab with a screwdriver; a stop limits the latch to 2.2 mm of travel so it can't be overbent, and the hook stays engaged even then. Rail contact is spread across the bracket's full width.



TPU CARD RETAINER (V0.4)

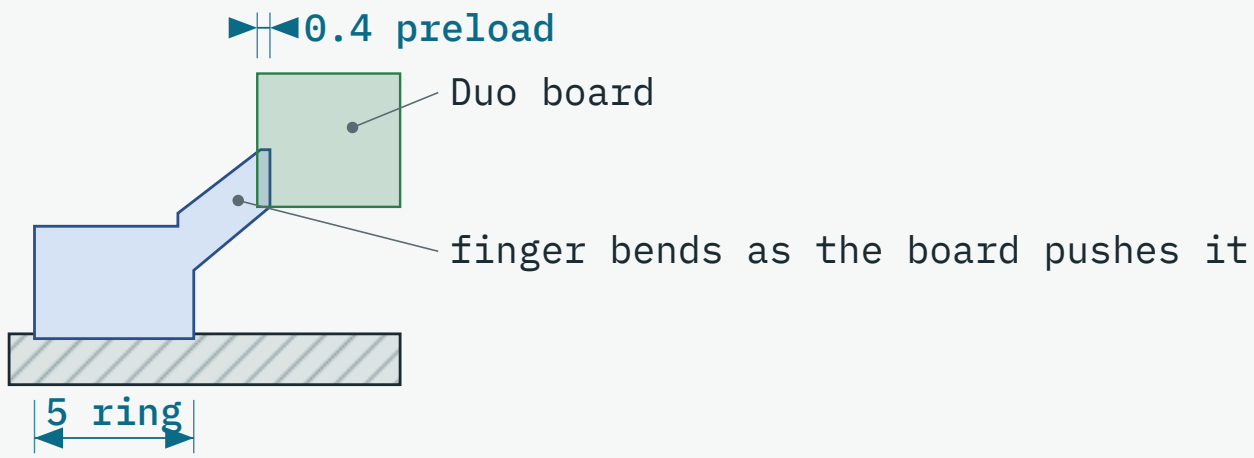
RETAINER · SEEN FROM THE MOUTH

TPU ring pushed into the mouth behind the boards, before the lid. It's a 0.15 mm press fit all round, and the middle stays open for the PCIe ribbon and air. Three leaning fingers spring against the board corners: both Duo corners and the Pi's GPIO-side corner. The Pi's power-button corner is left free. Four bumps drop into the first roof and floor vent slots of the printed body.



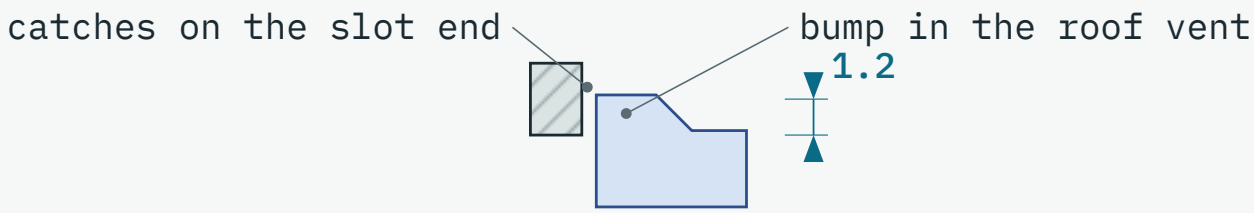
DETAIL · FINGER ON THE DUO CORNER (3:1)

Plan cut at the Duo. The finger leans about 40°, so the board edge bends it instead of crushing it: 0.4 mm of preload gives a few newtons per finger. Printed, the lean is a 40° overhang.



DETAIL · BUMP IN A ROOF VENT (3:1)

The bump's back face is square and catches on the mouth-side end of the vent slot, so the boards can't push the retainer out. It prints straight off the bed. To remove the retainer, press the four bumps in through the vents and pull it out by the ring.

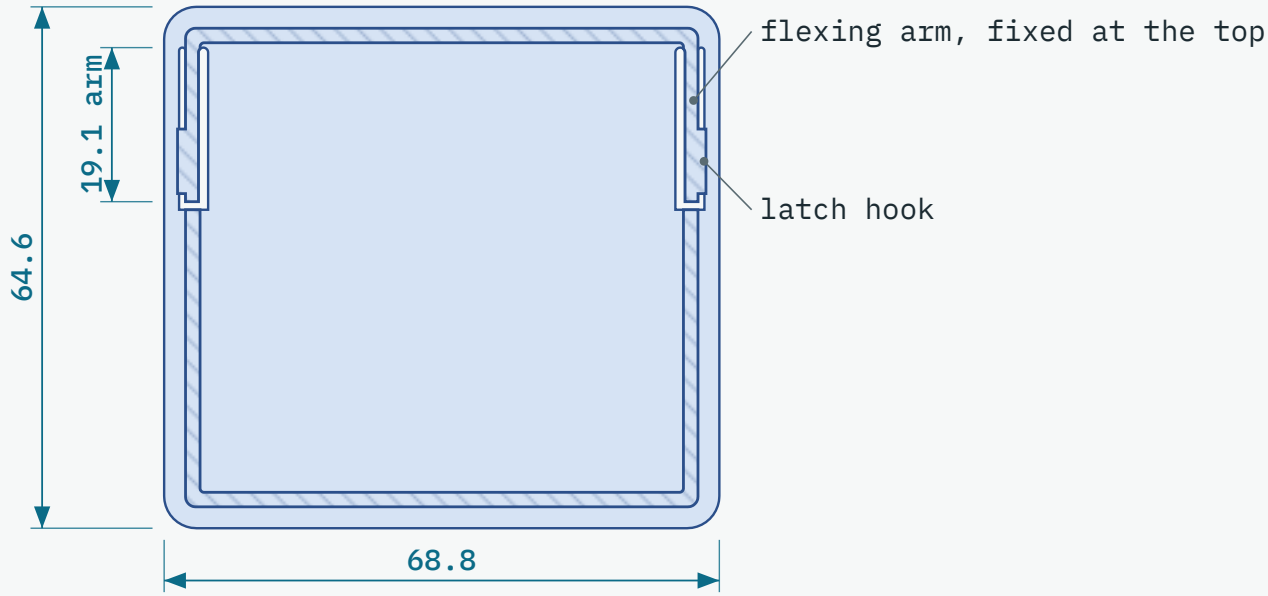


cut through a roof vent slot

LID AND FEET

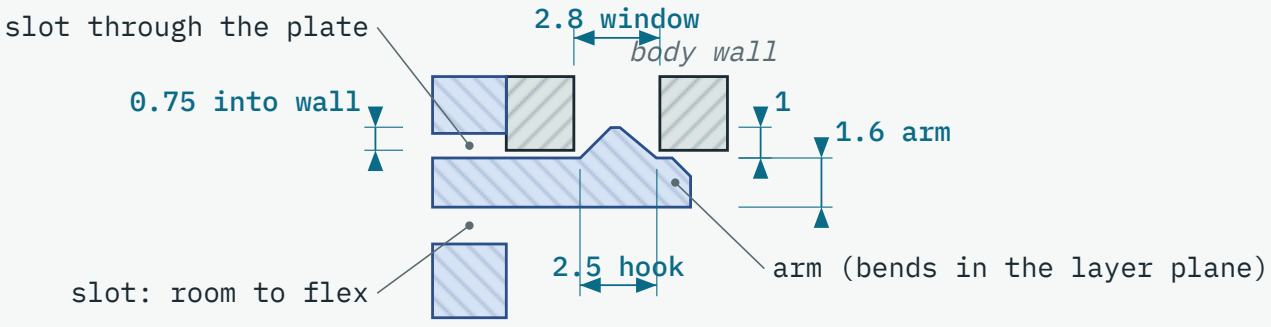
LID · INSIDE FACE

v0.4, PLA or PETG. A 6 mm skirt locates it in the mouth. On each long side, slots cut a 19 mm arm out of the skirt, fixed at its top end; its hook clicks into the body's upper side window. No prongs any more: the card retainer holds the boards, so the lid only closes the case.



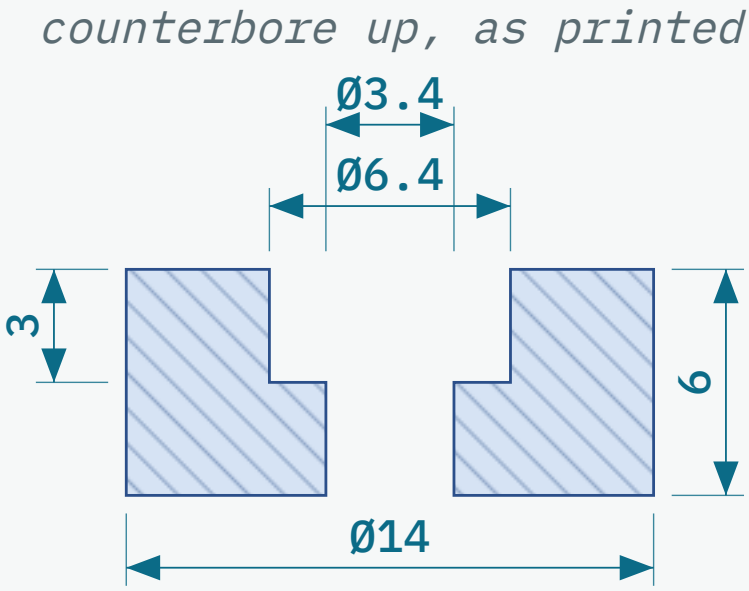
DETAIL · LATCH IN THE WINDOW (4:1)

Plan cut through the hook. The arm is 1.6 mm thick and bends sideways, in the plane of the print layers, so PLA flexes along its filaments; at the 0.75 mm it moves to pass the wall, the strain is about 0.9%. The hook's lower face is a 45° overhang as printed; the lead-in is gentler, so it pushes in more easily than it pulls out.



FOOT

v0.4. TPU, Ø14 × 6, printed counterbore-up. Fit it counterbore toward the desk: an M3 × 8 button or socket head sits in the Ø6.4 × 3 counterbore with 3 mm of TPU under it, and 5 mm of thread goes into the heat-set insert.



Design notes

WHERE THE SPACE GOES

VALUE	REASON
13 bay	9.5 mm for the SSDs on the Duo's underside (5 mm SSD standoffs plus a double-sided drive, with margin) and 3.5 mm of air above the floor vents.
12 gap	Duo top to Pi underside. The case sets it now that there are no standoffs; 12 mm is what Pimoroni's ribbon is made for. The upper lip (5 mm) sits in it with 7 mm clear above the Duo.
1 / 3 ledges	How far each ledge reaches under its board. The Pi's is short to stay clear of the solder joints on its underside.
16 header	Pi top to HAT underside, the usual height that clears the Active Cooler.
9 + 5	Tallest part on the HAT, then air under the roof vents.
4 sides	Air between the boards and the side walls, above the upper lip and in the recess between the lips, as you asked.
10 SD end	Room for the ribbon loop (about 7 mm) between the boards and the lid.
42 × 11	USB-C + HDMI window. A dock cable's USB-C overmold has about 8 mm of window either side and has to reach ~5 mm in to seat the plug.

ASSEMBLY

1. Fit the SSDs to the Duo and the ribbon to both boards as Pimoroni describe, but leave the standoffs out. Plug the HAT onto the Pi's GPIO and PoE headers. Fit your microSD card now if you use one.
2. Press four M3 × 5.7 heat-set inserts into the floor.
3. Fold the Duo under the Pi and slide both in from the open end, USB first: the Duo onto the lower ledges, the Pi onto the upper ledges. Push them together by their SD edges until they stop on the end wall.
4. Push the TPU card retainer in behind the boards, bumps up and down, until its four bumps drop into the first roof and floor vent slots. Its fingers press on the board corners.
5. Push the lid in until both latches click into the upper side windows.
6. Screw the TPU feet on with M3 × 8 button or socket heads.
7. To open, pull the lid out by the thumb notch at the top. If it's tight, press both hooks in through the side windows with a small screwdriver as you pull. The boards stay put. To take them out, press the retainer's bumps in through the vents and pull the ring.

PRINTING

- Body: USB end flat on the bed, PETG. Every wall, the lip and the drive-bay walls are vertical in that orientation. Vent slots and windows only bridge 3–10 mm. No supports, no brim.
- Lid: outer face on the bed, PLA (PETG also works and is more forgiving), 0.20 mm layers, no supports, no brim. The arm slots go right through the plate; check they're clear on the first layer. The hooks' undersides are 45° overhangs.
- Card retainer: TPU, back face on the bed. The bumps grow straight off the bed; the fingers lean about 40°.
- Feet: TPU, counterbore side up (no overhang). Turn them over to fit.
- The click is set by `latch_t` (1.6, arm thickness) and `latch_hook` (1). Too stiff: thin the arm to 1.2. Too loose: raise the hook to 1.2. Either way, reprint only the lid (48m).

HARDWARE

- 4 × M3 × 5.7 heat-set inserts, 4 × M3 × 8 button or socket head screws (feet)
- DIN mount instead of feet: 4 × M3 × 20 button or socket head screws, TS35 rail (35 × 7.5 or 35 × 15)
- Nothing for the boards. The Duo kit's standoffs and screws aren't used.

YOUR BRIEF (VERBATIM)

> Let’s make a raspberry pi 5 case. Requirements:
> - it has a PoE hat (check my Gmail inbox for order confirmations from Core Electronics for the part number)
> - it has a pimoroni dual nvme base
>
> I want to avoid screw holes, I would prefer. Refer to this sketch of the side profile showing how I would like the pi and the nvme base to sit on a lip
>
> The nvme base will need enough vertical clearance from the pi to account for thick nvme disks (dual Samsung 1TB)
>
> For the USBC port, it will need enough clearance to account for fairly chunky usb connectors like the ones on dell thunderbolt docks
>
> Previous designs I have made use. 3mm walls but that’s not a hard requirement. If strength can be gained through smart use of webbing and structural ribs then that would be ideal if print time can be optimised
>
> Printing in 2-3 parts would be ideal,
> - main body
> - lid
>
> If the lid is on the smallest face (the usb/rj45 end) or the sd card end that would be ideal. The pi can slide in like into the case as if it were a tube
>
> Attaching the end pieces “lids” can wither be screw in (m3, counter bore heads, or a nice click in place) for a screwless click in place I would print that from TPU to allow for some flex with whatever you use, and it can still retain some rigidity
>
> Give it ventilation holes top and bottom, you can increase the space between the inner walls and the nvme base/poe hat as they tend to generate a bit of heat
>
> It will also need some feet that can be screwed in with m3 counterbored screws
>
> Make some sort of tray for the as card so it can be slotted in and out like a SIM card
>
> I think those are all the requirements for now
>
> Create 2d projections before the 3d model

With a first sketch of the side profile: thin upper wall, a lip the board sits on, a thicker wall below.

Follow-ups:

> I'd prefer the cross section be like this [second sketch]
> I don't want to screw into the pi or the nvme base, the idea is for this to be simple
> Alright skip the sad card holder stuff
> 1. Samsung 980EVO
> 2. No idea
> 3. Not fussed, the gpio pins hold it in pretty snugly
> Let's create another component that uses the footprint of the feet to attach it to the case.
> It will be a bracket to mount it on a DIN rail
> Leave plenty of clearance for the airflow, make sure the bracket is sturdy and rigid as people may be rough installing and removing it
>
> 11 Nov, after printing:
> The case printed well, the little tabs on the lid did not print at all
> PLA might be able to do the job if you make the tabs thin enough to flex a little
> The rails are doing a good job of securing the nvme base
> I can shake the case and the base is secure
> Can you confirm your design of the feet suit these screws [2 photos: button-head screw in calipers]
> Then I think securing the cards should be separate from the back lid. So we make a TPU insert that locks the cards in tightly, then the pla lid goes on after that
> If there's a better way to do that then by all means suggest it and don't be 100% stuck to the shape of this bit [photo: sketch of a frame in the mouth pressing the board edges]

The second sketch: an upper lip with the Pi resting on top, a recess in the wall where the NVMe board's edge runs, a lower lip the NVMe board rests on, then the wall steps back out below.

Source: `pi5_case.scad` (OpenSCAD, every value above is a parameter) and a handoff README, saved in the Designing stuff project under `pi5-poe-nvme-case/`. Drawings are cut from the model's STLs.

Part data: PoE HAT (G) 64.98 × 56.5 · NVMe Base Duo kit (12 mm standoffs) · Duo assembly (SSDs face away from the Pi) · Pi 5 power button by the USB-C · Pi 5 mechanical drawing